

Spring 1993

Problem 1 (Sp93). Let (a_n) and (ε_n) be sequences of positive numbers. Assume that $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ and that there is a number k in $(0, 1)$ such that $a_{n+1} \leq ka_n + \varepsilon_n$ for every n . Prove that $\lim_{n \rightarrow \infty} a_n = 0$.

Problem 2 (Sp93). Let $A = (a_{ij})$ be an $n \times n$ matrix such that $\sum_{j=1}^n |a_{ij}| < 1$ for each i . Prove that $I - A$ is invertible.

Problem 3 (Sp93). Evaluate $\int_{-\infty}^{\infty} \frac{x^3 \sin x}{(1+x^2)^2} dx$.

Problem 4 (Sp82, Sp93). Suppose that the group G is generated by elements x and y that satisfy $x^5 y^3 = x^8 y^5 = 1$. Is G the trivial group?

Problem 5 (Sp93). Let k be a positive integer. For which values of the real number c does the differential equation

$$\frac{d^2 x}{dt^2} - 2c \frac{dx}{dt} + x = 0$$

have a solution satisfying $x(0) = x(2\pi k) = 0$?

Problem 6 (Sp77, Sp93, Fa94). Find a list of real matrices, as long as possible, such that

- the characteristic polynomial of each matrix is $(x-1)^5(x+1)$,
- the minimal polynomial of each matrix is $(x-1)^2(x+1)$,
- no two matrices in the list are similar to each other.

Problem 7 (Sp82, Sp93). Let $\{g_n\}$ be a sequence of twice differentiable functions on $[0, 1]$ such that $g_n(0) = g'_n(0) = 0$ for all n . Suppose also that $|g''_n(x)| \leq 1$ for all n and all $x \in [0, 1]$. Prove that there is a subsequence of $\{g_n\}$ which converges uniformly on $[0, 1]$.

Problem 8 (Sp93). Classify up to isomorphism all groups of order 45.

Problem 9 (Sp93, Sp05). Let a be a complex number and ε a positive number. Prove that the function $f(z) = \sin z + 1/z - a$ has infinitely many zeros in the strip $|\Im z| < \varepsilon$.

Problem 10 (Sp93). Let f be a real valued $\mathcal{C}^1$ function on $[0, \infty)$ such that the improper integral $\int_1^\infty |f'(x)| dx$ converges. Prove that the infinite series $\sum_{n=1}^\infty f(n)$ converges if and only if the integral $\int_1^\infty f(x) dx$ converges.

Problem 11 (Sp93). Let P be the vector space of polynomials over $\mathbb{R}$. Let the linear transformation $E : P \rightarrow P$ be defined by $Ef = f + f'$, where f' is the derivative of f . Prove that E is invertible.

Problem 12 (Sp93). Prove that for any fixed complex number ζ ,

$$\frac{1}{2\pi} \int_0^{2\pi} e^{2\zeta \cos \theta} d\theta = \sum_{n=0}^{\infty} \left(\frac{\zeta^n}{n!} \right)^2$$

Problem 13 (Su81, Sp93). Show that no commutative ring with identity has additive group isomorphic to $\mathbb{Q}/\mathbb{Z}$.

Problem 14 (Sp93). Prove that every solution $x(t)$ ($t \geq 0$) of the differential equation

$$\frac{dx}{dt} = x^2 - x^6$$

with $x(0) > 0$ satisfies $\lim_{t \rightarrow \infty} x(t) = 1$.

Problem 15 (Su78, Fa87, Sp93). Let R be the set of 2×2 matrices of the form

$$\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$$

where a, b are elements of a given field $\mathbb{F}$. Show that with the usual matrix operations, R is a commutative ring with identity. For which of the following fields $\mathbb{F}$ is R a field: $\mathbb{F} = \mathbb{Q}, \mathbb{C}, \mathbb{Z}_5, \mathbb{Z}_7$?

Problem 16 (Sp93). Prove that $\frac{x^2 + y^2}{4} \leq e^{x+y-2}$ for $x \geq 0, y \geq 0$.

Problem 17 (Sp82, Sp93). Prove that if G is a group containing no subgroup of index 2, then any subgroup of index 3 is normal.

Problem 18 (Sp93). Let f be an analytic function in the unit disc, $|z| < 1$.

1. Prove that there is a sequence (z_n) in the unit disc with $\lim_{n \rightarrow \infty} |z_n| = 1$ and $\lim_{n \rightarrow \infty} f(z_n)$ exists (finitely).
2. Assume f nonconstant. Prove that there are two sequences (z_n) and (w_n) in the disc such that $\lim_{n \rightarrow \infty} |z_n| = \lim_{n \rightarrow \infty} |w_n| = 1$, and such that both limits $\lim_{n \rightarrow \infty} f(z_n)$ and $\lim_{n \rightarrow \infty} f(w_n)$ exist (finitely) and are not equal.